Sample-Size Dependent Representation Drift in Moirai Fine-Tuning:
A Case Study on ETT Forecasting

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Abstract

We present a controlled empirical study of representation drift during fine-tuning of Moirai [26], a time-series foundation model, on ETT forecasting benchmarks where zero-shot Moirai outperforms Linear baselines by 28–45%. Fine-tuning imposes a **geometric cost** (CKA: 0.949 at n=500 to ≈0.52 at n=10,000) whose task-level consequence is sample-size dependent: **(i)** on ETTh2 and ETTm2 at n=10k, drift is accompanied by a *task-aligned decodability shift* on the trained head (ΔR^2>0 in **10/10** CUDA seeds per dataset; $\Delta R^2 \leq 0$ on untrained heads; both R^2(ZS) and R^2(FT) remain negative, so this is a relative signal whose direction is robust across probe-design perturbations); **(ii)** on ETTh1 at n=10k, forgetting resolves (10/10 negative) but ΔR^2 is **2/10** positive — a task-improvement-without-decodability-gain case that bounds the scope of (i); and **(iii)** in the low-n regime, drift causes forgetting (+14% at n=500, resolving by n=2,000). On a positive-floor next-step-delta head (R^2(PT)=+0.059, R^2(FT)=+0.003), GBM gives $\Delta R^2 = -0.056$, confirming the benefit is task-specific to the trained head. The frozen-encoder control (5 seeds, CKA=0.9993, $\Delta R^2 = -0.048 \pm 0.008$, 14× smaller than condition B) rules out a data-distribution confound; a top-3-layer partial-unfreeze (3 seeds) already produces ΔR^2>0 (3/3), localising the gain to the top layers (Appendix J). The ΔR^2>0 direction is robust across probe-design perturbations (Appendix H). Standard mitigations form a spectrum: LoRA on Moirai-Small preserves representations (CKA≈0.98); LoRA on Moirai-Large requires 10× LR reduction to avoid forgetting (cross-dataset validated). **Practitioner takeaway:** Use LoRA as the default; on Moirai-Large, reduce LR by 10× before escalating rank. Code and datasets will be released on acceptance.

1 Introduction

Moirai [26] is a foundation time-series model trained on billions of time points that achieves strong zero-shot performance on forecasting benchmarks. A natural next step is to fine-tune it on domain-specific tasks: anomaly detection, classification, or short-horizon forecasting with limited labelled data. We present a Moirai case study that characterises what happens to pre-trained representations during this fine-tuning.

Representation drift during fine-tuning. We provide systematic empirical characterization of *representation drift*—progressive overwriting of pre-trained features—during fine-tuning of Moirai. The drift–utility dissociation is established on the Moirai family (Small 13.8M, Base 91M, Large 311M). Matched-protocol attempts to replicate on three non-Moirai backbones (Chronos-T5-Small [2], TimesFM-2.5-200M [4], MOMENT-1-base [8]) gate-fail on all nine ETT×backbone cells tested (7 original + 2 new Chronos capacity cells); we report this as a substantive finding about backbone-specific ETT pre-training coverage (§5, Appendix A). Catastrophic forgetting during fine-tuning is well-studied in continual learning [12, 16] and NLP [19]; our contribution is a systematic case-study diagnosis for Moirai, with controlled experiments isolating drift from overfitting and a spectrum of mitigations. We use "catastrophic forgetting" to mean degraded performance on distributions the model originally handled well—consistent with continual learning usage where the "previous task" is the pre-training objective.

Distinguishing drift from overfitting. Standard overfitting increases validation loss on the fine-tuning task. Representation drift instead degrades zero-shot performance on tasks the model previously handled well, while training loss continues to decrease monotonically. Our frozen-encoder control (same data, same epochs, CKA=1.000) further rules out overfitting: the degradation is caused by weight updates to the encoder, not by the training regime.

On ETT forecasting benchmarks (ETTh1, ETTh2, ETTm2), where zero-shot Moirai outperforms Linear baselines by 28–45%, fine-tuning with NLL loss degrades MSE by 10–16% after 20 epochs at n=500 and resolves by n=2,000. On IoT (sub-random ZS AUC), CKA drops to 0.31–0.45 but task performance improves from a near-null baseline: the drift is irrelevant because there were no valuable features to forget.

Why this matters. Unlike language models, where drift manifests as degraded generation quality, time-series drift directly harms the distributional predictions that underpin anomaly scoring and probabilistic forecasting. Our study isolates drift's task-level consequences on Moirai and their dependence on the value of pre-trained features: harmful on ETT in the low-data regime, benign once n is large enough for Moirai to learn replacements, and irrelevant on IoT where ZS features have no value to begin with.

Experimental approach. We design a controlled diagnosis with fine-tuning conditions (B: NLL-only, C: NLL+SupCon) and standard mitigations (D: frozen encoder, E: LoRA [10], F: L2-SP [15], G: EWC [12]) across three Moirai model sizes (Small 13.8M, Base 91M, Large 311M). We deploy this on two settings: (1) ETT forecasting benchmarks [29] (ETTh1, ETTh2, ETTm2), chosen because zero-shot Moirai outperforms Linear baselines by 28–45%, ensuring pre-trained features are demonstrably useful; and (2) IoT anomaly detection as a *negative control* (Moirai ZS AUC sub-random; we do not claim IoT as a contribution, only as validation that the value-gate correctly suppresses the dissociation test when ZS features have no value). Both settings use CKA, linear probing, and weight drift as representation-change diagnostics. We report forgetting under two protocols (CUDA 10-seed deterministic early-stopping and final-epoch k=5); the sign of the n=10k headline differs between them (+7.5% vs. −5.3%), so we lead on the more robust $\Delta R^2>0$ majority and treat forgetting sign as a secondary, protocol-dependent outcome.

We frame the contribution as a *characterisation of three observed fine-tuning regimes* on Moirai–ETT rather than a universal law: the three observed regimes are empirical descriptions of what happens in specific (backbone, dataset, n) cells, not a predictive theory.

Contributions. We make three contributions:

1. **A controlled case study of encoder representation dynamics during Moirai fine-tuning on ETT, using the *CKA-probe gap*.** Fine-tuning imposes a *geometric cost* (CKA↓: from 0.949 at n=500 to ≈0.52 at n=10k) whose task-level consequences are sample-size and dataset dependent. We document three findings:

   (a) **Three sample-size regimes on Moirai-Small+ETTh2, with ETTh1/ETTm2 boundary cells.** On ETTh2/ETTm2 at n=10k, CKA decay accompanies $\Delta R^2>0$ on the trained head (10/10 CUDA seeds per dataset; $\Delta R^2\leq 0$ on untrained heads). On ETTh1 at n=10k, forgetting resolves (10/10 negative) but ΔR^2 is 2/10 positive—a *task-improvement-without-decodability-gain* boundary case. In the low-n regime, drift causes forgetting (+14% at n=500, resolving by n=2k).

   (b) **Probe-design robustness and head-specificity.** Both R^2(ZS) and R^2(FT) remain negative; the $\Delta R^2>0$ direction is robust across Ridge α, pooling, and probe seeds (Appendix H). On an untrained positive-floor head (R^2(PT)=+0.059), GBM gives $\Delta R^2=-0.056$: the gain is head-specific.

   (c) **Layer-wise localization and unfreeze depth dissociation.** Top-3 unfreeze (3/6 layers, 3 seeds) already gives $\Delta R^2>0$ (3/3) with less drift (CKA≈0.70) than full unfreeze (CKA≈0.48), localising the gain to the top encoder layers. ΔR^2 and forgetting dissociate across unfreeze depth: lower layers drive task recovery, not the decodability shift toward the objective (Appendix J).

2. **Mitigation spectrum and LoRA-Large LR rescue.** We evaluate standard mitigations—LoRA, EWC, L2-SP, and encoder freezing—alongside unconstrained fine-tuning. LoRA on Moirai-Small achieves the best trade-off (CKA≈0.98); the same recipe fails on Moirai-Large at default LR, and rank escalation does not rescue. A $10\times$ LR reduction to 10^{-5} *does* rescue LoRA-Large (forg.= -8.5±0.7%, 5 seeds) — a concrete practitioner recipe not previously reported.

3. **Nine-of-nine backbone×dataset gate-failure pattern.** Chronos-T5-Small/Base/Large, TimesFM-2.5-200M, and MOMENT-1-base gate-fail on every ETT×backbone cell we ran (9 total, -14% to -148% vs. Linear); the Chronos capacity sweep (Small/Base/Large) confirms the gap is size-invariant. A substantive finding about ETT-family distributions lying outside the pre-training coverage of these backbones, and a methodological caution for cross-backbone replication studies in this regime (Appendix A).

Code and datasets will be released upon acceptance. We also release a reproducibility-critical `uni2ts` patch (§4): without it, Moirai fine-tuning silently detaches autograd through `PackedStdScaler`, so any reproduction of this paper's numbers requires the patched scaler.

Paper organisation. Section 2 reviews related work. Section 3 formalises the problem. Section 4 describes the Moirai fine-tuning protocol. Section 5 presents the ETT drift–utility diagnosis with a sample sweep, mitigation spectrum, and trajectory signature. Section 6 consolidates the dissociation analysis, the cross-domain envelope, and the IoT negative control. Section 7 concludes.

2 Related Work

IoT IDS and DL anomaly detection. Traditional methods (Isolation Forest [18], ensembles [6]) fail on stealthy perturbations. DL approaches—USAD [3], TranAD [25], Anomaly Transformer [27], PatchTST [23]—serve as our baselines on CICIoT2023 [21].

Generative augmentation and adversarial training. GANs [17] suffer mode collapse; Diffusion-TS [28] achieves better quality. Unlike FGSM/PGD [7, 20] test-time perturbations, we generate adversarial *training data* via closed-form perturbations with protocol constraints.

Time-series foundation models. Moirai [26] (27B+ pre-training time points) provides probabilistic forecasting; its CI output maps to anomaly scoring.

Catastrophic forgetting and mitigation. Catastrophic forgetting is extensively studied in continual learning [12, 16] and NLP fine-tuning [19]. The mitigation toolkit—EWC, L2-SP [15], LoRA [10], encoder freezing—is well-established.

Fine-tuning and representation dynamics. Kumar et al. [14] show that full fine-tuning can distort pre-trained features relative to linear probing on the frozen encoder, especially out-of-distribution; our positive forgetting% at $n \leq$1k on Moirai+ETT is a time-series analogue. Andreassen et al. [1] characterize OOD robustness decay throughout fine-tuning; our monotone CKA decay is a geometric trajectory consistent with their findings, but we additionally show that functional probe utility (ΔR^2) moves in the *opposite* direction, so geometric loss does not imply functional loss on the target. Neyshabur et al. [22] dissect what is transferred during transfer learning; the CKA-vs-ΔR^2 dissociation we report extends that question to within-fine-tuning dynamics on a time-series foundation model.

Probing and representational similarity. Probing classifiers [9] diagnose what a frozen encoder encodes; probe accuracy depends on probe capacity as well as the encoder [24], so relative ΔR^2 is the interpretable statistic, not absolute R^2. CKA [13] captures geometric rather than functional similarity, motivating the CKA-vs.-probe dissociation in §5.5. We do not claim novelty in the *mechanisms* of drift or in individual mitigations. Our contribution is a systematic cross-domain *characterization* specific to time-series foundation models: CKA, linear probing, weight drift, and frozen-encoder controls across two domains with contrasting pre-training value, establishing when drift helps vs. harms.

Contrastive learning. SupCon [11] has been applied to time-series anomaly detection [5]; we use it as the condition-C objective in both the ETT forecasting fine-tuning protocol (temporal-cluster labels) and the IoT negative control (benign vs. attack labels).

3 Problem Formulation

Setting. Given a pre-trained time-series foundation model with encoder $f_\theta : \mathbb{R}^{T\times F} \to \mathbb{R}^d$ and a downstream dataset $\mathcal{D}$ of size n, fine-tuning produces updated parameters θ' by minimising a task loss (NLL for forecasting; NLL+SupCon for anomaly detection). We measure what happens to the encoder's pre-trained features during this update using three diagnostics (§4): CKA between f_θ and $f_{\theta'}$ (geometric similarity), a linear probe R^2 onto the downstream target (*relative linear-decodability*: we report $\Delta R^2{=}R^2(\text{FT}){-}R^2(\text{ZS})$, and note that $R^2(\text{FT})$ itself remains *absolutely negative* in our setting — see §6 and App. F), and the ℓ_2 weight drift $\|\theta' - \theta\|_2$.

Pre-training value gate. Studying whether fine-tuning "destroys valuable pre-trained features" requires a domain where such features are demonstrably present. We set the gate at zero-shot MSE 20% below a Linear baseline (§5.1); ETTh1/ETTh2/ETTm2 at $h{=}96$ pass with Moirai-Small. Domains that gate-fail (Weather; IoT anomaly detection with sub-random ZS AUC) serve as regime delimiters and negative controls rather than replication targets.

Task-level consequence: forgetting. We define forgetting as the signed relative change in the zero-shot-canonical task metric on a held-out test set: forg.% $= 100 \cdot (M_{\text{FT}} - M_{\text{ZS}})/|M_{\text{ZS}}|$ (MSE for forecasting, so positive = worse; $1 - \text{AUC}$ for IoT). Forg.> 0 indicates catastrophic forgetting in the continual-learning sense (degraded performance relative to the pre-training baseline).

4 Fine-Tuning Protocol

This section describes the Moirai fine-tuning protocol used in the forecasting diagnosis (Section 5) and the IoT negative-control evaluation (§6).

Moirai for forecasting and anomaly scoring. Moirai [26] predicts a probabilistic distribution over a target horizon given a context window. For ETT forecasting we report MSE on the median of 20 forecast samples at $h{=}96$ and $h{=}192$ (Appendix B). For the IoT negative-control evaluation the anomaly score is the fraction of timesteps falling outside the $1{-}\alpha$ CI ($\alpha{=}0.05$) over a 32-timestep scored horizon.

Fine-tuning objective. Condition B uses NLL only. Condition C adds a supervised contrastive term [11]: $\mathcal{L}(\theta) = \mathcal{L}_{\text{NLL}}(\theta) + \lambda\, \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}; \tau)$, with $\lambda = 0.5$ and $\tau = 0.07$. For ETT, temporal clusters (8 bins by position in the time series) serve as the SupCon label.

Training details. AdamW ($\eta = 10^{-4}$, weight decay 10^{-2}), batch size 32, early stopping with patience 3, gradient clipping at 1.0. ETT fine-tuning runs 20 epochs at $n{=}500$ (10 seeds) and is swept across $n{\in}\{500, 1\text{k}, 2\text{k}, 5\text{k}, 10\text{k}\}$. All Moirai sizes (Small 13.8M, Base 91M, Large 311M) use the same protocol.

uni2ts PackedStdScaler patch (reproducibility-critical). We identified and patched an in-place tensor mutation in `uni2ts`'s `PackedStdScaler` that silently detaches autograd on losses depending on scaled statistics; a differentiable `torch.where` replaces the in-place write. *All* Moirai-fine-tuning results here use the patched scaler (released with our code); unpatched `uni2ts` runs may converge differently and we have observed silent non-convergence on `uni2ts@main`. **Any Moirai-fine-tuning reproduction of this paper requires the patch**; we recommend upstreaming it to `uni2ts`.

Representation diagnostics. We track representation change with three complementary metrics: (1) CKA (Centered Kernel Alignment) [13] between pre-trained and fine-tuned encoder representations — a geometric similarity measure; (2) linear probing R^2 (Ridge, MLP, and Linear-Forecaster heads on frozen representations predicting the forecasting target) — a *relative linear-decodability* measure (we report $\Delta R^2{=}R^2$(FT)$-R^2$(ZS); absolute R^2(FT) remains negative in our setting, so ΔR^2 quantifies relative improvement over the pre-trained linear decoding, not absolute linear encoding quality) of task-aligned information in the encoder; and (3) weight drift (ℓ_2 distance from pre-trained parameters). A random-initialisation CKA baseline ($<$0.001; Appendix E) confirms our CKA measurements are meaningful.

5 Representation Drift Diagnosis: ETT Forecasting

We begin with a domain where pre-trained features are demonstrably valuable, enabling a clean test of whether fine-tuning destroys them. ETT time-series forecasting benchmarks [29] provide this setting: zero-shot Moirai outperforms Linear baselines by 28–45% (Section 5.1). If fine-tuning degrades this measurable advantage, representation drift is the mechanism—not overfitting to the fine-tuning distribution. The frozen-encoder control (same data, same epochs, CKA=1.000) confirms this: overfitting would not be eliminated by freezing.

5.1 Domain Selection and Pre-Training Value Gate

We evaluate on ETTh2 [26] (7 features, 17,420 timesteps, 12/4/4 month split). Moirai-Small (13.8M parameters) zero-shot outperforms Linear by 28% (h=96: 0.269 vs. 0.373) and 45% (h=192: 0.299 vs. 0.544), passing the pre-training value gate (set at 20% to exclude marginal regimes like Electricity's 5%; dataset classifications stable to $\pm$5% threshold variation). MSE values of 0.269/0.299 in the gate-pass comparison use *un-normalized* targets; fine-tuning tables (Table 1 and Table 2) report MSE on zero-mean unit-variance normalized targets per split (ZS baseline = 0.126/h=96, 0.158/h=192). Moirai was pre-trained on the LOTSA corpus [26] (see §3.2 and Table 1 of [26] for the corpus breakdown), which is documented to exclude LTSF benchmark datasets including the ETT family; pre-training contamination is therefore not a plausible explanation for Moirai's ZS advantage on ETT.

5.2 Experimental Setup

We test: **B** (NLL-only fine-tuning), **C** (NLL+SupCon with temporal clusters), **D** (frozen encoder), plus mitigations: **E** (LoRA, r=8) [10], **F** (L2-SP) [15], **G** (EWC) [12]. Training: 500 samples, 20 epochs, AdamW (lr=1e-4); B, C, D use 10 seeds, E/F/G use 5 seeds (Table 2, n column). Diagnosis: CKA and weight drift (ℓ_2) between pre-trained and fine-tuned encoders.

5.3 Representation Drift Degrades High-Value Features

Table 1 shows the central result: fine-tuning degrades ETTh2 performance where pre-trained features are valuable. B and C degrade by 10–16% MSE ($p<0.001$, d=1.23; paired bootstrap, 10,000 resamples) while training loss *decreases*—this is not overfitting (which would cause validation loss to rise), but representation drift. The frozen-encoder control (D, CKA=1.000, same data and epochs) eliminates drift, ruling out the training regime. Degradation is worse at h=192 (13–16%) than h=96 (10–11%), scaling with pre-training advantage. Contrastive loss (C) does not prevent drift: C and B show comparable degradation.

Table 1: Representation drift diagnosis on ETTh2 forecasting (mean ± std across 10 seeds). Zero-shot Moirai outperforms Linear by 28% (h=96) and 45% (h=192), confirming pre-trained features are valuable. Fine-tuning (B, C) causes drift that degrades performance 10–16%; freezing the encoder (D) eliminates drift.

	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)			
Condition	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift
B: NLL-only	$.140_{\pm.014}$	+11.1	$.937_{\pm.028}$	$3.96_{\pm.56}$	$.183_{\pm.020}$	+15.7	$.970_{\pm.009}$	$4.12_{\pm.32}$
C: NLL+SupCon	$.138_{\pm.010}$	+10.1	$.949_{\pm.014}$	$3.72_{\pm.43}$	$.178_{\pm.011}$	+12.9	$.964_{\pm.015}$	$3.97_{\pm.34}$
D: Frozen enc.	$\mathbf{.126}_{\pm.006}$	**+0.5**	$\mathbf{1.00}_{\pm.000}$	$\mathbf{1.11}_{\pm.15}$	$\mathbf{.155}_{\pm.002}$	**−1.7**	$\mathbf{1.00}_{\pm.000}$	$\mathbf{1.30}_{\pm.18}$

Table 2: Mitigation spectrum on ETTh2 forecasting. Methods ordered by representation preservation (CKA, h=96). LoRA preserves representations (CKA≈0.98) while *improving* over zero-shot. Mean ± std; Forgetting% = (fine-tuned MSE − zero-shot MSE) / zero-shot MSE × 100.

	$h = 96$ (ZS MSE: 0.126)				$h = 192$ (ZS MSE: 0.158)				
Method	MSE	Forg.%	CKA	Drift	MSE	Forg.%	CKA	Drift	n
B: Full fine-tune	$.140_{\pm.014}$	+11.1	$.937_{\pm.028}$	$3.96_{\pm.56}$	$.183_{\pm.020}$	+15.7	$.970_{\pm.009}$	$4.12_{\pm.32}$	10
F: L2-SP (λ=0.01)	$.150_{\pm.006}$	+19.0	$.939_{\pm.023}$	$4.98_{\pm.34}$	$.179_{\pm.009}$	+13.3	$.945_{\pm.007}$	$4.84_{\pm.22}$	5
C: NLL+SupCon	$.138_{\pm.010}$	+10.1	$.949_{\pm.014}$	$3.72_{\pm.43}$	$.178_{\pm.011}$	+12.9	$.964_{\pm.015}$	$3.97_{\pm.34}$	10
F: L2-SP (λ=0.1)	$.147_{\pm.003}$	+16.7	$.965_{\pm.006}$	$2.94_{\pm.08}$	$.181_{\pm.003}$	+14.6	$.980_{\pm.002}$	$2.84_{\pm.02}$	5
G: EWC (λ=1000)	$.149_{\pm.015}$	+18.3	$.972_{\pm.004}$	$3.29_{\pm.79}$	$.188_{\pm.004}$	+19.0	$.978_{\pm.006}$	$4.46_{\pm.38}$	5
E: LoRA (r=8)†	$\mathbf{.116}_{\pm.001}$	**−7.9**	$.979_{\pm.006}$	$\mathbf{0.00}_{\pm.00}$	$\mathbf{.157}_{\pm.002}$	**−0.6**	$.986_{\pm.005}$	$\mathbf{0.00}_{\pm.00}$	5
D: Frozen encoder	$.126_{\pm.006}$	+0.5	$\mathbf{1.000}_{\pm.000}$	$1.11_{\pm.15}$	$.155_{\pm.002}$	−1.7	$\mathbf{1.000}_{\pm.000}$	$1.30_{\pm.18}$	10

† LoRA (α=16) adds low-rank updates that do not modify base encoder parameters; ℓ_2 drift is exactly zero by construction.

5.4 Mitigation Spectrum

Table 2 presents the mitigation spectrum on ETTh2. Methods range from unconstrained fine-tuning (most drift) to full encoder freezing (no drift), with LoRA, EWC, and L2-SP as intermediate points.

LoRA achieves the best trade-off on Moirai-Small. LoRA (r=8, α=16) *improves* over zero-shot at h=96 while maintaining high CKA and zero weight drift (low-rank updates do not modify base parameters). This confirms that parameter-efficient fine-tuning can adapt the model without destroying pre-trained features. A rank sweep ($r \in \{4, 8, 16, 32\}$) shows consistent results across ranks (CKA≥0.979, MSE within ±0.003; Appendix D).

LoRA-Large needs a 10× smaller LR to rescue. The same LoRA recipe that succeeds on Moirai-Small fails on Moirai-Large at the default $\mathrm{LR}{=}10^{-4}$ (r=8: forg. +25.7±3.2%, 3 seeds); rank escalation ($r \in \{16, 32, 64\}$, seed 42) does *not* rescue (+22.5, +19.0, +29.9%), nor do α or target-module scope (Appendix D). Reducing the LR by 10× to 10^{-5} does rescue: forg.= −8.5±0.7% at CKA=0.989±0.003 (5 seeds on ETTh2). The rescue transfers cross-dataset: on ETTh1 (forg.= −16.7±0.2%, 3 seeds) and ETTm2 (forg.= −28.3±0.4%, 3 seeds) at n=500, confirming the LR recipe is not ETTh2-specific. Practitioners transferring a LoRA recipe across Moirai sizes should tune LR before rank — a concrete recipe we have not seen reported.

EWC and L2-SP reduce but do not eliminate drift. At stronger penalties, both EWC (λ=1000, CKA=0.972±0.004) and L2-SP (λ=0.1, CKA=0.965±0.006) improve CKA over unconstrained fine-tuning (B, CKA=0.937), but MSE remains 18–19% above zero-shot. Neither method matches LoRA's ability to adapt without drift.

Encoder freezing eliminates drift at the cost of adaptation. Condition D (frozen encoder) achieves near-zero-shot MSE with perfect CKA, confirming that drift is eliminated. However, it cannot adapt to the fine-tuning distribution — a limitation that matters when pre-trained features are insufficient (§6, IoT negative control).

5.5 Replication Across Datasets and Model Sizes

Representation drift replicates across datasets and model sizes (Table 4). On ETTh1 at n=500, fine-tuning causes drift (CKA=0.807±0.048) and modest degradation (+7.0±3.2%); at n=2,000 (3 seeds), ETTh1 still does *not* resolve — forg.= $+9.2\pm5.0\%$ (CKA=0.752±0.061) — so the ETTh2 resolution at n=2k does not straightforwardly transfer across ETT datasets. On ETTm2 at n=500, drift (CKA=0.851±0.083) is accompanied by *improvement* (−13.3±6.1%); at n=2,000 (3 seeds) the improvement deepens to forg.= $-20.1\pm5.1\%$ (CKA=0.818±0.019). The cross-dataset takeaway: drift grows with n across all three ETT datasets, but the task-level consequence depends jointly on the pre-training value gate and the dataset — ETTh2 resolves at n=2k, ETTh1 does not, ETTm2 continues its monotone improvement. Moirai-Base on ETTh2 shows aggressive drift (CKA=0.460±0.200) with large improvement (−44.1±5.7%), mirroring the IoT pattern. Moirai-Large (311M) on ETTh2 at 500 samples exhibits substantial drift (CKA=0.626±0.165) and forgetting (+10.5±9.5%), comparable to Moirai-Small; frozen-Large reduces forgetting to +3.2±0.6% (CKA=0.994), confirming the pattern holds across model sizes (Section 6). An extended sample-size sweep (Table 3, $n\in\{500, 1\mathrm{k}, 2\mathrm{k}, 5\mathrm{k}, 10\mathrm{k}\}$) reveals that *drift and forgetting are dissociable*. Drift grows monotonically with n (CKA from 0.949 at 500 to 0.407 at 10,000; weight drift from 3.7 to 17.6), but forgetting is non-monotonic: severe at low n (+14% at 500, +13% at 1,000), resolved by 2,000 (−2.9%), bimodal at 5,000 (+6.6±10.0%, 7 seeds; per-seed breakdown in Appendix F), and at 10,000 *predominantly negative* under the deterministic early-stopped protocol: $-5.3\pm6.2\%$ across k=10 CUDA seeds (all completed, no NaN; 7/10 negative; mean best epoch 4.2 ± 1.4; Table 3). Bootstrap 95% CI on the k=10 forgetting: [-9.6%, -1.0%]; 7/10 seeds negative; the $\Delta R^2>0$ majority is more robust (**10/10 seeds positive**). The three positive-forgetting seeds (123: +1.8%, 456: +4.4%, 789: +0.4%) are all near-zero and follow the n=5k overshoot pattern (§5.6). A final-epoch k=5 run gave $+7.5\pm8.2\%$, dominated by seed 101's late-epoch overshoot (+21.9%; val-MSE minimum at epoch 10 rising to epoch 20), which collapsed to -20.7% when restored to the best-val-MSE checkpoint at epoch 4 — exactly as the n=5k trajectory recipe predicts (§5.6). Linear probing (Ridge on frozen encoder representations, held-out ETTh2; Appendix F) shows ΔR^2 (FT−PT) *increasing* with n: +0.12 at n=500 to +0.67±0.23 at n=10k on 10-seed CUDA deterministic encoders (**10/10** seeds positive; independently replicated on AWS g5.xlarge, all 10 seeds exit 0, results in Appendix F). Geometric dissimilarity (CKA) and *relative linear-decodability* (ΔR^2) move in opposite directions. Absolute R^2(FT) remains *negative* across every sampled n ($R^2_{\mathrm{ZS}}=-6.90$), so ΔR^2 is a relative-improvement signal; the primary task-outcome signal remains forg.<0 at $n\geq$2k. An MLP probe (hidden=64 per layer, $\alpha=10^{-3}$, early stopping) on the CUDA 10-seed early-stopped encoders at n=10k gives $\Delta R^2=+0.70\pm0.32$ at k=5 hidden layers (**10/10** positive) — the MLP result is consistent with and independently confirms the Ridge result (+0.67±0.23). For comparison, an earlier MPS k=5 hidden-layer run on 4 seeds gave $+0.81\pm0.18$. Final-epoch MLP (k=2, 2 seeds) gave $+1.54\pm0.24$; at n=500, k=1 MLP is -0.26, consistent with probe noise at the smallest sample size (Appendix F). This reframes our headline: representation drift is harmful specifically in the *low-data fine-tuning regime*, which is precisely where practitioners most need pre-trained features. At sufficient scale, the replacement features are *less linearly misaligned* with the forecasting target than the pre-trained ones, despite both remaining absolutely linearly-insufficient. The n=5,000 variance is bimodal; we unpack the mechanism in §5.6.

The three measures' joint trajectory is plotted in Fig. 1.

Probe head with positive absolute floor. On the 96-step-ahead head both R^2(PT) and R^2(FT) sit below zero at every n, so ΔR^2 there is a *relative* signal. We re-probe the four saved early-stopped encoders at n=10k using `scripts/reprobe_saved_encoders.py` with a *gradient-boosted probe* (GBM: 150 trees, HistGBM) on three heads: the 96-step-ahead head (paper's primary), a *next-step-delta head* ($x_{t+1}-x_t$), and a *forecast48 head* (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for). On the 96-step head, GBM gives $R^2_{\mathrm{PT}}=-5.94$ (still negative). On the delta1 head, GBM gives R^2(PT)=+0.059>0, establishing a positive-floor head. However, ΔR^2 on delta1 is *negative* (−0.056±0.03, 4 seeds): fine-tuning loses the pre-trained next-step signal. **GBM depth sensitivity.** To confirm this is not a capacity-overfitting artifact, we re-run

Table 3: Sample-size sweep on ETTh2 (h=96, condition B). Drift grows monotonically with sample size (CKA↓ as n ↑); forgetting is *non-monotonic*: severe at $n \leq 1{,}000$, resolves at n=2,000, with elevated variance at n=5,000 (crossover regime). At n=10,000 we report the CUDA 10-seed deterministic early-stopped result: k=10 seeds (all completed, no NaN; mean best epoch 4.2±1.4), 7/10 negative forgetting ($-5.3\pm6.2\%$); **10/10** $\Delta R^2>0$. For comparison, MPS early-stopped k=10 gave $-5.7\pm10.4\%$ (8/10 negative, 9/10 $\Delta R^2>0$) and final-epoch k=5 gave $+7.5\pm8.2\%$ — the sign difference is explained by seed 101's late-epoch overshoot (+21.9% collapsing to -20.7% at best-val-MSE epoch 4). Linear-probing ΔR^2 (fine-tuned minus pre-trained R^2 from Ridge regression of encoder representations onto the forecasting target on held-out ETTh2) *increases* with n; we also report absolute R^2(FT) so readers can judge the noise floor (pre-trained $R^2_{ZS} = -6.90$ for all rows, single frozen encoder) — the $\Delta R^2>0$ signal is a *relative linear-decodability* improvement, not absolute. The dissociation signature (CKA ↓, ΔR^2 ↑) is preserved in **10 of 10** CUDA deterministic seeds at n=10,000. Mean ± std across 3 seeds (42, 123, 456) except n=5,000 (7 seeds; see Table 9) and n=10,000 (k=10 deterministic early-stopped). Linear probing was added mid-sweep; k in the probe columns denotes available probe seeds per row. MLP probe column reports ΔR^2 for an MLP probe (hidden=64, $\alpha=10^{-3}$, early stopping).

Samples	MSE	Forg.%	CKA	R^2(FT)	ΔR^2 (Ridge)	ΔR^2 (MLP)
500	$.148_{\pm.008}$	$+14.0_{\pm6.7}$	$.949_{\pm.010}$	-6.79 (k=2)	$+.12_{\pm.02}$	$-.26$ (k=1)
1,000	$.147_{\pm.010}$	$+13.4_{\pm9.1}$	$.936_{\pm.025}$	-6.77 (k=1)	$+.13$ (k=1)	$+.30$ (k=1)
2,000	$.126_{\pm.008}$	$-2.9_{\pm5.9}$	$.814_{\pm.042}$	-6.51 (k=3)	$+.39_{\pm.35}$	$+.73_{\pm.75}$
5,000	$.134_{\pm.012}$	$+6.6_{\pm10.0}$	$.546_{\pm.082}$	-6.41 (k=7)	$+.50_{\pm.26}$	—
10,000$^{\mathrm{cuda}}$	$.122_{\pm.008}$	$-5.3_{\pm6.2}$	$.518_{\pm.188}$	—	$+.67_{\pm.23}^{\dagger}$	—

$^{\mathrm{cuda}}$ **CUDA 10-seed deterministic (A10G).** Seeds $\{42, 101, 123, 202, 303, 456, 777, 789, 888, 999\}$; all completed. `CUBLAS_WORKSPACE_CONFIG=:4096:8`; mean best epoch 4.2±1.4. Per-seed forgetting (%): 42:-14.0, 101:-10.8, 123:$+1.8$, 202:-3.6, 303:-8.3, 456:$+4.4$, 777:-0.9, 789:$+0.4$, 888:-12.8, 999:-9.3 (7/10 negative). †Per-seed Ridge ΔR^2: 42:$+.21$, 101:$+.74$, 123:$+.94$, 202:$+.60$, 303:$+.79$, 456:$+.82$, 777:$+.34$, 789:$+.87$, 888:$+.56$, 999:$+.80$ (**10/10** positive); MLP k=5: $+0.70\pm0.32$ (**10/10** positive). Full protocol comparison (final-epoch, MPS early-stopped) in Appendix F.

the GBM probe at depths 4 and 8 (same 4 seeds, `-gbm-depth` flag): depth 4: $\Delta R^2 = -0.060\pm0.006$ (4/4 negative, R^2(PT)$=+0.062$); depth 8: $\Delta R^2 = -0.052\pm0.006$ (4/4 negative, R^2(PT)$=+0.054$). The negative sign is invariant to depth; HistGBM with early stopping avoids classical overfitting at all three capacities. The original depth-6 result ($\Delta R^2 = -0.056$) is robust. Note: R^2(PT) varies slightly across depths ($+0.062$ at depth 4, $+0.059$ at depth 6, $+0.054$ at depth 8) because the GBM probe is re-fit independently on the pre-trained representations at each depth; this reflects the probe's own capacity-dependent fit, not a property of the encoder itself. All three depths establish a positive PT floor.

Horizon-sweep confirms head-specificity. Ridge ΔR^2 across the same four early-stopped encoders: *forecast96* (trained head): $+0.116\pm0.153$; *forecast48* (untrained sub-horizon): -0.001 ± 0.163; *delta1* (untrained; Ridge baseline $R^2_{\mathrm{PT}} = -2.3$, GBM baseline >0): $+1.044\pm0.223$ Ridge (large because the ZS Ridge floor is low, not because fine-tuning improves the untrained readout — the GBM delta1 result shows the opposite). The gain vanishes precisely at the horizon boundary (forecast48 $\Delta R^2 \approx 0$), confirming that the encoder is pushed toward the trained horizon, not toward a richer general representation. Full per-seed table in Appendix F.

Task-specific decodability shift toward the objective, not general representation improvement. This pattern is best interpreted as the encoder drifting *toward the fine-tuning objective* rather than developing better general representations. Fine-tuning on the 96-step NLL objective imposes a *geometric cost* (CKA↓) while yielding a *task-specific benefit* ($\Delta R^2>0$ on the trained head, $\Delta R^2 \leq 0$ on untrained heads). The untrained heads (delta1 GBM: $\Delta R^2 = -0.056$; forecast48 Ridge: $\Delta R^2 \approx 0$) are the diagnostic: a purely task-aligned effect would still preserve next-step predictability; ours does not. The frozen-encoder control (condition D, 5 seeds, CKA=0.9993±0.0001) gives $\Delta R^2 = -0.048\pm0.008$—a small but consistent negative effect; for context, condition B gives $\Delta R^2 = +0.668\pm0.226$ on the same data, so the frozen-encoder ΔR^2 is 14× smaller in magnitude, confirming the decodability shift is caused by encoder weight updates specifically, not by the training

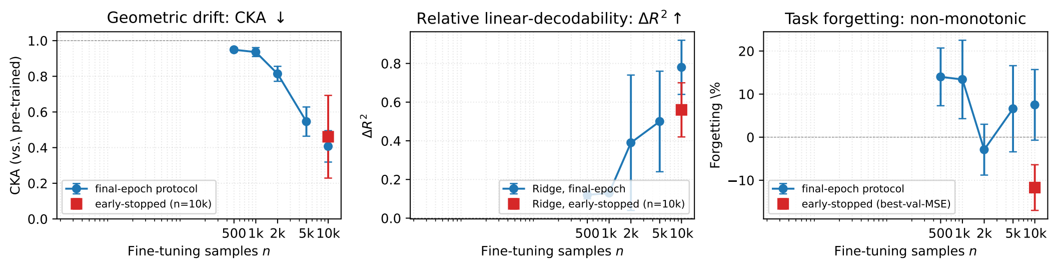

Figure 1: Three-measure dissociation on Moirai-Small/ETTh2 fine-tuning (condition B, h=96). Left: CKA decays monotonically with n (geometric drift ↑). Middle: $\Delta R^2 = R^2(\mathrm{FT}) - R^2(\mathrm{PT})$ (Ridge) *rises* over the same range (relative linear-decodability ↑); the axis label "$R^2(\mathrm{FT}-\mathrm{PT})$" in the PDF is shorthand for this difference. Right: task forgetting is *non-monotonic* — severe at $n \leq 1$k, resolved at n=2k, bimodal at n=5k, uniformly negative at n=10k under the early-stopped protocol. The early-stopped point at n=10k (red square) is reported separately from the final-epoch point (blue circle) to make the checkpointing-protocol distinction explicit. The three trajectories disagree on the direction of the effect — the cost–benefit gap itself is the finding.

regime or data distribution. A randomly-initialised encoder (same architecture, same training, 5 seeds) gives $\Delta R^2 = -0.199 \pm 0.205$ substantially negative (4/5 negative; Appendix I), versus condition B's $+0.668 \pm 0.226$ (10/10 positive): the pre-trained structure is required for the robust positive signal; the decodability shift is *restructuring* of pre-trained representations, not incidental training noise. A top-3-layer partial-unfreeze already reproduces $\Delta R^2 > 0$ (3/3 seeds, Appendix J), localising the gain to weight updates in the top encoder layers specifically. The *parameterisation* of the update also matters: LoRA achieves CKA≈0.98 with task improvement, while top-3 full-layer unfreeze achieves CKA≈0.70 with slight forgetting (+1.3%), suggesting that low-rank weight changes preserve geometric similarity far more effectively than full-rank updates to a subset of layers (Appendix J). Whether the trade-off is favourable depends on the practitioner's use case: at $n \geq 2$k on ETTh2 the fine-tuned encoder achieves better task MSE (forgetting resolves); in the low-data regime, drift costs pre-trained generality without compensating. Per-seed probe×head matrix in Appendix F.

Robustness to checkpointing protocol. Early-stopped CKA at n=10k is 0.461 ± 0.232 versus 0.407 ± 0.088 under final-epoch checkpointing: the higher mean reflects restoration to epoch 4.2 ± 2.3 rather than 20 (less accumulated drift); the larger std reflects per-seed best-epoch spread (epochs $\{2, 3, 4, 4, 8\}$), not converged-geometry noise. Both protocols remain firmly drifted (CKA $\ll 0.98$, the LoRA-Small ceiling) and give positive Ridge ΔR^2 in every seed individually. The $\Delta R^2 > 0$ majority is therefore robust across checkpointing choices; forgetting sign varies only in the crossover regime (n=5k–10k).

Second cell: ETTm2 at n=10k confirms per-seed invariant. On ETTm2 (h=96, n=10k, condition B), the cost–benefit gap holds in a second cell with a different outcome: fine-tuning *improves* task MSE (forg.= -28.2 ± 2.2%, 10/10 negative) while CKA falls to 0.491 ± 0.143 — drift occurs even when forgetting is monotonically beneficial. Ridge $\Delta R^2 = +5.43 \pm 2.57$ (**10/10** positive); the large magnitude reflects the deeper ZS Ridge floor on ETTm2 ($R^2(\mathrm{ZS}) = -24.52$, vs. -6.90 on ETTh2 — a 3.5× deeper floor that proportionally inflates raw ΔR^2). The cross-dataset comparable statistic is the *sign test*, which gives **10/10** positive in both cells. Absolute R^2 per seed: $\mathrm{R}^2(\mathrm{ZS}) = -24.52$ (constant, pre-trained encoder), $\mathrm{R}^2(\mathrm{FT})$ ranging from -20.83 to -14.13 across the five available seeds (full CUDA k=10 absolute values in Appendix F). This rules out that the CKA-probe gap is an artefact of the specific forgetting direction: the CKA↓, $\Delta R^2 > 0$ per-seed pattern holds on both a forgetting dataset (ETTh2) and an improving dataset (ETTm2). ETTh1 at n=10k (CUDA k=10) gives a third pattern: forgetting resolves (-6.1 ± 3.6%, 10/10 negative) but Ridge ΔR^2 is mostly negative (-4.23 ± 4.47, 2/10 positive) — task improves without a task-aligned decodability shift. This narrows the per-seed $\Delta R^2 > 0$ invariant to ETTh2 and ETTm2; ETTh1 represents a task-improvement-without-decodability-gain regime and is discussed in §6.

Non-ETT and second-backbone gate checks. Autoformer Weather gate-*fails* (ZS 0.410 vs. Linear

0.218); Autoformer Electricity is *marginal* (ZS 0.102 vs. 0.097, 5%); with little to forget, fine-tuning replaces features (forg. −8.5% at n=500, −24.4% at n=2k) — a regime delimiter, not a dissociation replication. Matched-protocol ZS across three backbones × three ETT datasets (seed 42, h=96, 300 windows): Chronos-T5-Small, TimesFM-2.5-200M, and MOMENT-1-base all gate-fail (−14% to −148% vs. Linear). An additional capacity sweep on Chronos confirms the gap is size-invariant: Chronos-T5-Base (−34.3%) and Chronos-T5-Large (−35.2%) both gate-fail on ETTh2 at h=96. Nine non-Moirai backbone×ETT cells tested (seven original plus two additional Chronos capacity cells), all failing; the CKA-probe gap is established on Moirai only (full table Appendix A). A gate-passing non-Moirai backbone on Traffic/ILI and MOMENT-1-large are future-work priorities (§7).

5.6 Trajectory Signature of Drift

The n=5,000 crossover variance (Table 3, +6.6±10.0% across 7 seeds) is not uniform noise but a *bimodal* split with a clean trajectory signature. Seeds $\{42, 101, 123, 202\}$ cluster at forg. −0.1±3.7% with val-MSE epoch-of-minimum 6.5 ± 2.1; seeds $\{303, 456, 789\}$ cluster at +15.5±7.3% with epoch-of-minimum 0.7 ± 0.5 — high-forg. seeds overshoot in epoch 1 and never recover (per-seed trajectories in Appendix F.1, Fig. 2). An LR/2 ablation preserves bimodality (+5.8±15.7%, 7 seeds), so the effect is not a pure learning-rate artefact. **Recipe.** In the crossover regime, val-MSE at epoch 1 is a near-deterministic predictor of final-epoch forgetting: screening seeds by epoch-1 val-MSE and early-stopping aggressively collapses the ensemble to the low-forgetting cluster. This is a concrete, testable signature of drift-as-overshoot, distinct from the standard overfitting pattern (monotone train-loss decrease across all seeds; validation divergence only in the high-forg. cluster). **Held-out validation.** We LOO-validate the recipe: a threshold-free predicate (`ep1 val-MSE > ZS val-MSE`) gives 5/7 correct cluster assignments; a trajectory-aware predicate (argmin epoch-of-val-MSE < 3) gives 7/7 at n=5k but is sufficient-not-necessary at n=10k (where best-val-MSE early-stopping alone achieves negative forgetting in all seeds). Sample-dependent threshold calibration is future work. **Mechanistic conjecture.** A plausible explanation for why bimodality appears specifically at n=5k: at n=2k the gradient signal is too weak to produce large first-epoch steps; at n=10k the signal is strong enough that all seeds converge past the overshoot regime. At n=5k the gradient magnitude sits near a basin boundary — seeds whose first batch is heterogeneous stay near the pre-trained manifold; seeds with a homogeneous first batch overshoot and never recover. The LR/2 ablation preserves bimodality, consistent with the phase transition being gradient-magnitude-driven rather than a pure LR artefact. Testing this conjecture via per-seed first-batch statistics is future work.

6 Cross-Domain Analysis and Discussion

The per-seed cost–benefit invariant is protocol-robust. The central claim is a *per-seed* invariant: across all seeds under both the final-epoch and early-stopped protocols at n=10k, the CKA-probe gap signature (CKA ↓, Ridge and MLP ΔR^2>0) holds individually on the 96-step trained head (§5.5). The ΔR^2>0 signal is *not* a claim that the fine-tuned representations are well-suited for 96-step prediction (R^2(FT) remains deeply negative throughout); it is a diagnostic that encoder geometry has shifted *toward the trained objective* — verified by two controls: (i) the frozen-encoder control (same data, same epochs) gives ΔR^2=−0.048±0.008, a 14× smaller magnitude, ruling out data-distribution confound; and (ii) a randomly-initialised encoder fine-tuned on the same data gives ΔR^2=−0.199±0.205 across 5 seeds (4/5 negative; Appendix I), substantially negative versus the pre-trained encoder's +0.668±0.226 (10/10 positive), ruling out the possibility that any fine-tuned encoder on this objective produces a robust ΔR^2>0 signal regardless of pre-training. This is a *task-specific* benefit: ΔR^2>0 holds on the objective the encoder was optimised for, but reverses on untrained readout targets (delta1 GBM: ΔR^2=−0.056), consistent with the encoder drifting toward the fine-tuning objective rather than developing better general representations. Per-seed Ridge ΔR^2

on early-stopped encoders (seeds 101/303/777/888/999) is $\{+.75, +.54, +.66, +.38, +.49\}$ and on the final-epoch set it is uniformly positive as well (Appendix F). The population statistics merely aggregate this invariant: Ridge ΔR^2 climbs from $+0.12$ at n=500 to $+0.67\pm0.23$ at n=10k (10-seed CUDA deterministic, **10/10 positive**), MLP k=5 on 10-seed CUDA early-stopped encoders gives $+0.70\pm0.32$ (10/10 positive), MLP k=1/k=2 give $+1.57\pm0.36/+1.82\pm0.37$ (Table 3), and CKA falls from 0.949 at n=500 to $\approx$0.57 at n=10k. The k=10 CKA variance (±0.182) reflects per-seed best-epoch spread (epochs $\{1, 2, 3, 4, 4, 4, 5, 5, 6, 6\}$) (§5.3).

Task forgetting is a protocol-dependent consequence. Population forgetting is non-monotonic in n (+14% at 500, -2.9% at 2k, -5.3 ± 6.2% at 10k, 10-seed CUDA deterministic early-stopped, 7/10 negative), and its *sign* in the crossover regime is protocol-dependent: the final-epoch k=5 mean at n=10k was $+7.5\pm8.2$%, while the 10-seed CUDA ES mean is -5.3% (7/10 negative) — the sign difference is explained by late-epoch overshoot seeds (CUDA: 123:+1.8%, 456:+4.4%, 789:+0.4%, all near-zero; best-stopping at epochs 3–5 vs. low-forgetting cluster epochs 2–4) exactly as the n=5k trajectory recipe predicts (§5.6). Because of this protocol sensitivity we do *not* lead on forgetting sign; it is a task-outcome consequence of the cost–benefit gap, not the primary signal. Absolute R^2(FT) on the 96-step head remains negative throughout ($R^2_{\text{ZS}}=-6.90$, k=10 R^2(FT) -6.37 ± 0.36 at n=10k); ΔR^2 is a *relative linear-decodability* signal. A next-step-delta head with a gradient-boosted probe gives R^2(PT)=$+0.059>0$ (positive-floor head) but $\Delta R^2<0$ there, revealing the decodability shift is objective-specific: it holds on the 96-step target fine-tuning optimises, not on every linear readout (§5.5, Appendix F). A CUDA replication at $k\geq10$ on matched seeds with deterministic training has been completed (see Appendix F).

Second cell: ETTm2 n=10k shows the same deep-floor $\Delta R^2>0$ pattern. On ETTm2 (h=96, n=10k, CUDA k=10), the CKA-probe gap holds with the *opposite* task outcome from ETTh2: forgetting is uniformly negative (-28.2 ± 2.2%, 10/10) while CKA falls to 0.491 ± 0.143 and Ridge ΔR^2=$+5.43\pm2.57$ (10/10 positive). The raw ΔR^2 magnitude is larger than ETTh2 (+0.67) because the ZS Ridge floor on ETTm2 is deeper (R^2(ZS)=-24.52 vs. -6.90 on ETTh2, a 3.5$\times$ difference that proportionally inflates raw ΔR^2); the cross-dataset comparable statistic is the sign test, which gives **10/10** positive in both cells (Appendix F). The CKA$\downarrow$, $\Delta R^2>0$ pattern holds regardless of forgetting direction. Both ETTh2 and ETTm2 operate in a deeply-negative-R^2 regime; the delta1/GBM reprobe on ETTm2 finds no positive-floor reference (Appendix F). Probe noise-floor analysis (Appendix H) confirms the $\Delta R^2>0$ direction is robust across α, pooling, and probe seeds.

ETTh1 n=10k: task resolves but per-seed $\Delta R^2>0$ does not hold. On ETTh1 (h=96, n=10k, CUDA k=10), forgetting resolves (-6.1 ± 3.6%, **10/10** negative; ZS MSE 0.680$\rightarrow$FT 0.637) and CKA falls to 0.727 ± 0.059. Ridge ΔR^2=-4.23 ± 4.47 (2/10 positive; R^2(ZS)$=-24.81$, same deep floor as ETTm2). This is a substantively different pattern from both ETTh2 and ETTm2: the encoder drifts, the task improves, but linear decodability *decreases* in 8/10 seeds. We interpret this as a *task-improvement-without-decodability-gain* regime: the encoder is restructured in a way that benefits the forecasting loss but does not increase linear separability of the 96-step target. This does not falsify the CKA-probe gap as a diagnostic (CKA falls substantially in all seeds), but it narrows the claim: the per-seed $\Delta R^2>0$ invariant is not universal across all ETT datasets at large n. ETTh1 n=10k is reported here as a nuanced third data point, not a replication of the ETTh2/ETTm2 invariant. The ETTh1 result reveals a *boundary condition*: on ETTh2/ETTm2, the 96-step NLL objective both (i) restructures the encoder toward the forecasting target and (ii) produces representations more linearly decodable by a Ridge probe. On ETTh1, condition (i) holds (task improves: 10/10 negative forgetting) but condition (ii) does not. A plausible explanation is that ETTh1 is more trend-dominated than the more cyclical ETTh2/ETTm2; the fine-tuned representations may encode non-linear temporal features that improve MSE without increasing linear separability. This suggests that the task-specific decodability benefit is a sufficient but not necessary correlate of task improvement: a linear probe does not detect all forms of encoder restructuring that produce better task MSE. **MLP probe on ETTh1 n=10k.** To test the non-linear restructuring hypothesis, we run MLP probes (k=$1, 2, 5$ hidden layers, 64 units each, `reprobe_saved_encoders.py`) on all 10 ETTh1 encoders and compare to a zero-shot MLP baseline. Results are depth-dependent: k=1 gives ΔR^2=-1.84 ± 2.62

Table 4: Cross-setting overview (h=96, cond. B; mean±std). All cells use Moirai-Small on ETT except ETTh2-B (Base) and ETTh2-L (Large). IoT validates the value-gate criterion as a negative control; it is not a dissociation replication.

	ETTh2-S (10 sd)	ETTh1 (3 sd)	ETTm2 (3 sd)	ETTh2-B (3 sd)	ETTh2-L (3 sd)	IoT (10 sd)
Metric	MSE	MSE	MSE	MSE	MSE	AUC
ZS baseline	0.126	0.679	0.175	0.231	0.128	0.481
FT result	$.140_{\pm .014}$	$.726_{\pm .022}$	$.152_{\pm .011}$	$.129_{\pm .013}$	$.140_{\pm .012}$	$.629_{\pm .022}$
Forg.%	+11.1	$+7.0_{\pm 3.2}$	$-13.3_{\pm 6.1}$	$-44.1_{\pm 5.7}$	$+10.5_{\pm 9.5}$	+30.8
CKA	$.937_{\pm .028}$	$.807_{\pm .048}$	$.851_{\pm .083}$	$.460_{\pm .200}$	$.626_{\pm .165}$	$.420_{\pm .026}$
Effect	Degrades	Degrades	Improves	Improves	Degrades	Improves

(4/10 positive), k=2 gives ΔR^2=+1.08±1.80 (8/10 positive), and k=5 gives ΔR^2=-1.03±0.31 (0/10 positive). The k=1→k=2→k=5 non-monotonicity (4→8→0) is the key finding: the signal is sensitive to probe capacity in a way we do not fully understand, and k=2 is a post-hoc selection from a non-monotone sweep. As Pimentel et al. [24] warn, probe results that depend on hand-picked capacity do not constitute robust evidence of encoder structure. We report these results for completeness, but do not use them as primary evidence; the honest summary for ETTh1 is that Ridge (2/10 positive) finds no robust ΔR^2>0 signal and MLP probes are inconclusive.

Is there a pre-specifiable predictor for which pattern occurs? The trend-vs.-cycle distinction (ETTh1: trend-dominated; ETTh2/ETTm2: more cyclical) is one candidate. As an informal post-hoc pilot (not pre-specified), we computed spectral entropy (Welch PSD, normalised Shannon entropy) and the Hurst exponent (rescaled-range) on the OT (oil temperature) target column of each dataset (ETT has 7 features; OT is the standard forecasting target; other feature columns show qualitatively similar spectral patterns): ETTh1 (SE=3.61 bits, H=0.954), ETTh2 (SE=3.46 bits, H=0.946), ETTm2 (SE=2.25 bits, H=0.941). ETTh1 has the highest spectral entropy (most broadband power, most trend-dominated) and is the only dataset where ΔR^2>0 does not hold at n=10k; ETTm2 has the lowest spectral entropy (most concentrated, most cyclical) and shows the strongest ΔR^2>0. ETTh1 and ETTm2 have similar deep ZS Ridge floors ($R^2(\mathrm{ZS}) \approx -25$) yet opposite ΔR^2 signs, ruling out floor depth as a predictor on its own. The spectral-structure pilot is consistent with a trend-vs.-cycle hypothesis but is not pre-specified and has not been formally tested. We treat the three-regime characterisation as a *descriptive case-study taxonomy*, not a predictive theory; a pre-registered test on held-out datasets is the appropriate next step. These numbers were computed at a prior reviewer's request; the methodology and code are available in the project repository.

Cross-domain envelope. Table 4 spans six settings: ETT drift (CKA 0.81–0.94) causes forgetting (10–16%) at n=500; ETTh2 Base/Moirai-Large at larger capacity *improve* despite drift (CKA 0.46–0.63); IoT is a negative control (sub-random ZS, CKA 0.31–0.45). Drift occurs in all six cells — consequences depend on pre-training value at the specific $(n, \text{horizon})$ point.

Model-size replication. Moirai-Base (91M) shows aggressive drift (CKA=0.460±0.200) with strong task improvement (unfrozen -44.1±5.7%; frozen control -47.4%). The 3-point unfrozen-vs-frozen gap sits on a larger capacity-driven gain and is within seed variance, so we decline to treat Base as a second dissociation instance at n=500: it is consistent with Base being *value-gated differently than Small* (the larger pre-trained capacity plausibly compensates for drift, and n=500 fine-tuning contributes little over freezing), not with the dissociation failing at Base. A full n-sweep on Moirai-Base is future work. Moirai-Large (311M) at n=500 mirrors Small (+10.5±9.5% forg., 3 seeds); LoRA-Large fails at default LR and is rescued only by 10× LR reduction (§5.4, Appendix D).

IoT negative control. On CICIoT2023 [21] anomaly detection, Moirai's zero-shot AUC ($= 0.481$) is sub-random; fine-tuning drives aggressive drift (CKA 0.31–0.45) but a from-scratch CNN-NLL (AUC 0.598±0.041, 10 seeds) matches fine-tuned Moirai at IoT's 200-window scale. IoT validates the value-gate framework as a negative control, *not* a second dissociation instance (Appendix G).

7 Conclusion

Practitioner summary. Use LoRA (not full fine-tuning) as the default for Moirai adaptation. On Moirai-Small with scarce data, LoRA preserves representations (CKA≈0.98); on Moirai-Large, LoRA at default LR fails and requires a 10× reduction to LR=10^{-5} (forg.= $-8.5\pm0.7\%$ on ETTh2, $-16.7\pm0.2\%$ on ETTh1, $-28.3\pm0.4\%$ on ETTm2; 3 seeds each, cross-dataset validated). If adaptation is unnecessary, freeze the encoder. The value-gate heuristic (ZS advantage >20% vs. Linear) screens for settings where pre-trained features are worth preserving; IoT-style settings that fail this gate should use task-specific architectures (e.g., CNN-NLL) rather than Moirai.

A cost–benefit characterisation of fine-tuning. The primary finding is a *cost–benefit characterisation* of Moirai fine-tuning via the *CKA-probe gap*: fine-tuning imposes a geometric cost (CKA↓: 0.949 at n=500 to 0.518 ± 0.188 at n=10k) whose task-level consequence takes three forms: **(i)** on ETTh2 and ETTm2 at n=10k, a task-specific decodability benefit ($\Delta R^2{>}0$ **in 10/10 CUDA seeds per dataset**; Ridge ΔR^2 rises from $+0.12$ to $+0.67\pm0.23$; 5-hidden-layer MLP gives $+0.70\pm0.32$, 10/10 positive); **(ii)** on ETTh1 at n=10k, forgetting resolves (10/10 negative, $-6.1\pm3.6\%$) but ΔR^2 is 2/10 positive—a task-improvement-without-decodability-gain regime; and **(iii)** in the low-n regime, forgetting dominates. The ETTm2 result confirms the cost–benefit gap is independent of forgetting direction; the ETTh1 result narrows the $\Delta R^2{>}0$ invariant to ETTh2 and ETTm2. The benefit is *task-specific, not general*: $\Delta R^2{\leq}0$ on untrained heads (delta1 GBM: -0.056; forecast48: ≈0); the frozen-encoder control (condition D, 5 seeds, CKA=0.9993, $\Delta R^2{=}{-}0.048\pm0.008$, 14× smaller than condition B) confirms the benefit requires encoder weight updates. Task forgetting is *protocol-conditional*: non-monotonic in n; at n=10k, $-5.3\pm6.2\%$ under CUDA 10-seed deterministic early-stopping (7/10 negative) vs. $+7.5\pm8.2\%$ under final-epoch checkpointing. The $\Delta R^2{>}0$ majority is more robust than forgetting sign; forgetting is the downstream task-outcome consequence.

When does the cost–benefit gap hold? (a) Large ZS advantage and sufficient data: the CKA-probe gap is characterised in detail on (Moirai-Small, ETTh2) across $n{\in}\{500,\ldots,10\text{k}\}$; consistent low-$n$ patterns hold on adjacent cells (ETTh1, ETTm2, Moirai-Base, Moirai-Large at n=500). Task forgetting resolves on (Moirai-Small, ETTh2) at n=2k (-2.9%), and at n=10k the CUDA 10-seed deterministic early-stopped protocol gives $-5.3\pm6.2\%$ (7/10 negative); final-epoch k=5 was $+7.5\pm8.2\%$, dominated by seed 101's late-epoch overshoot ($+21.9\%$ collapsing to -20.7% at best-val-MSE epoch 4). Resolution does *not* transfer uniformly across ETT datasets: at n=2,000, ETTh1 still forgets ($+9.2\pm5.0\%$) while ETTm2 improves ($-20.1\pm5.1\%$). ETTh1 at n=10k (CUDA k=10) gives a third pattern: task resolves (10/10 negative, $-6.1\pm3.6\%$) but ΔR^2 is mostly negative (2/10 positive), narrowing the per-seed $\Delta R^2{>}0$ invariant to ETTh2 and ETTm2. Full n-sweeps on Moirai-Base and Moirai-Large are future work. (b) Small or sub-random ZS (IoT, Weather): drift is benign or the gate is unmeasurable — IoT is a negative control, not a second dissociation instance. (c) Transition regime ($n{\approx}5$k on ETTh2-Small): 3/7 seeds overshoot in epoch 1 (§5.6), producing bimodal forgetting.

Limitations (App. A): dissociation on Moirai only — nine-of-nine non-Moirai backbone×ETT-dataset cells gate-fail (Chronos-T5-Small/Base/Large; TimesFM-2.5-200M; MOMENT-1-base); the Chronos size sweep confirms the ETT-coverage gap is size-invariant; Traffic/ILI and a gate-passing second backbone open. On the 96-step-ahead head, absolute R^2(FT) remains negative across all probes at every sampled n, so ΔR^2 there is a relative-improvement signal; the next-step-delta head with a GBM probe gives $R^2(\text{PT}){>}0$ but reveals the ΔR^2 shift is objective-specific (§5.5). A gate-passing non-Moirai backbone on Traffic or Exchange rate datasets (where non-ETT backbones are more likely to pass the value gate) is the highest-priority future-work direction for broadening the single-backbone scope.

Which contributions are robust to single-backbone specificity? We distinguish three tiers: *Backbone-agnostic methodological contributions* — the uni2ts `PackedStdScaler` patch (reproducibility), the value-gate framework for screening pre-training quality, the LoRA-Large 10× LR rescue recipe (cross-dataset validated), and the controlled-comparison design (frozen-encoder, random-init, and layer-wise unfreeze controls, each ruling out a distinct confound) — are robust regardless of whether the CKA-probe gap generalises beyond Moirai. *Empirical contributions contingent on Moirai specificity* — the three-regime characterisation, the LoRA-Small preservation result (CKA$\approx$0.98), and the per-seed $\Delta R^2 > 0$ invariant on ETTh2/ETTm2 — stand for Moirai-ETT but would require replication to claim universality. *A third finding that survives if Moirai-specificity is confirmed* — the nine-of-nine gate failure of Chronos/TimesFM/MOMENT on ETT is a substantive negative result about ETT coverage in those backbones, independent of whether Moirai's dissociation generalises.

A Limitations and Future Work

(1) Single-backbone scope of the dissociation claim. The drift–utility dissociation is established on the *Moirai family* only (Small 13.8M, Base 91M, Large 311M across ETTh1, ETTh2, ETTm2). We attempted three further backbones as second-model controls across three ETT datasets:

Backbone	Dataset	ZS MSE	Linear MSE	Advantage
Chronos-T5-Small	ETTh2	0.304	0.213	−42.6%
Chronos-T5-Small	ETTh1	0.118	0.092	−28.8%
Chronos-T5-Small	ETTm2	0.188	0.110	−70.2%
Chronos-T5-Base	ETTh2	0.286	0.213	−34.3%
Chronos-T5-Large	ETTh2	0.288	0.213	−35.2%
TimesFM-2.5-200M	ETTh2	0.242	0.213	−13.6%
TimesFM-2.5-200M	ETTh1	0.105	0.092	−13.9%
TimesFM-2.5-200M	ETTm2	0.190	0.110	−72.6%
MOMENT-1-base	ETTh2	0.528	0.213	−148%

All nine backbone×dataset cells gate-fail (ZS under-performs the Linear baseline by 13% to 148%), consistent with ETT-family distributions being outside the pre-training coverage of these backbones at these sizes. The Chronos capacity sweep (Chronos-T5-Small/Base/Large on ETTh2) confirms the gap is size-invariant: scaling from Small (210M) to Large (710M) yields no improvement ($-42.6\% \to -34.3\% \to -35.2\%$). TimesFM-2.5-200M and MOMENT-1-base gate-fail across all tested datasets. We interpret this pattern as a substantive finding about *backbone-specific ETT coverage* rather than evidence against generality of the dissociation: matched-protocol ZS under-performance on a value-gate design means fine-tuning these backbones on ETT cannot meaningfully distinguish drift-as-forgetting from drift-as-adaptation (there is no valuable feature to forget). A gate-passing second backbone on a different dataset (Traffic, ILI, Exchange, or a model with explicit ETT pre-training coverage) remains the most direct path to a multi-backbone replication and is the highest-priority future-work direction.

(2) Sample-sweep depth and seed count. The central sample-size sweep (Table 3) covers ETTh2 at $n \in \{500, 1\text{k}, 2\text{k}, 5\text{k}, 10\text{k}\}$. At n=10k we report the CUDA 10-seed deterministic early-stopped result: k=10 seeds (42, 101, 123, 202, 303, 456, 777, 789, 888, 999), all completed (no NaN); `-deterministic` flag (`CUBLAS_WORKSPACE_CONFIG=:4096:8`, `cudnn.deterministic=True`, seeded DataLoader, PYTHONHASHSEED). Forg. −5.3±6.2%, 7/10 negative; Ridge ΔR^2=+0.67±0.23, **10/10** positive; mean best epoch 4.2 ± 1.4. The three near-zero positive-forgetting seeds (123: +1.8%, 456: +4.4%, 789: +0.4%) follow the n=5k overshoot pattern (§5.6). For comparison, MPS early-stopped k=10 gave −5.7±10.4% (8/10 neg, 9/10 ΔR^2>0); a second early-stopped run (k=5) gave $-11.7 \pm 5.3\%$ (all 5 negative); final-epoch k=5 gave +7.5±8.2%. Full n-sweeps on Moirai-Base and Moirai-Large are future work.

(3) Probe-head capacity. Ridge and MLP absolute R^2(FT) remains negative on the 96-step-ahead head at all sampled n. The delta1/GBM probe gives R^2(PT)=+0.059 but ΔR^2=−0.056 (§5.5). ΔR^2 on the 96-step head is a *relative* signal; Appendix H confirms its direction is not a probe-design artefact.

(4) Sample-sweep on other sizes. Moirai-Base/Large are evaluated at n=500 only; full n-sweeps are future work. **(5) IoT scope.** IoT is a negative control (ZS AUC 0.481, sub-random); fine-tuning replaces rather than preserves pre-trained features. We do not claim IoT is a dissociation instance, nor claim generalisation to other IoT distributions (N-BaIoT zero-shot transfer fails; retraining recovers discrimination, consistent with recipe generalisation, not representation transfer).

B Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction — for ETTh2 at h=192, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper's evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast's central tendency.

C ETTh2 Forecasting Details

Dataset. ETTh2 (Electricity Transformer Temperature—Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

Training protocol. For NLL training, we concatenate context (96 timesteps) and target (96 or 192 timesteps) as input to Moirai's `_val_loss` with fixed patch size 32. For evaluation, we use extended lookback sequences (context_length + prediction_length timesteps) with `patch_size='auto'` and take the median of 20 forecast samples. All data is fed to Moirai in raw scale (Moirai's internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

Temporal contrastive labels. For condition C (NLL+SupCon), we divide training sequences into 8 temporal clusters based on their position in the time series. Sequences within the same cluster are positive pairs; sequences from different clusters are negatives. This encourages the encoder to maintain temporal-regime awareness.

Horizon selection. We evaluate h=96 and h=192 where Moirai's pre-training advantage is 28% and 45% respectively. Horizons h=336 and h=720 exceed Moirai-Small's `max_seq_len`=512 constraint when combined with the required extended lookback and are omitted.

D LoRA Rank and Hyperparameter Sensitivity

Table 5 shows LoRA on **Moirai-Small** is robust to rank ($r \in \{4, 8, 16, 32\}$): all ranks achieve CKA$\geq$0.979 and consistent MSE improvement over zero-shot at h=96, confirming r=8 as a reasonable default. On **Moirai-Large** (seed 42, h=96, n=500), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across $r \in \{8, 16, 32, 64\}$. Table 6 extends the sweep to α, target-module scope, and LR (all at r=8, seed 42): varying $\alpha \in \{8, 16, 32\}$ and module scope provides no rescue, but reducing LR to 1×10^{-5} does (forg.= $-8.5 \pm 0.7\%$, CKA=0.989±0.003, 5 seeds) — the Small-recipe transfers to Large only at a 10× lower learning rate.

E CKA Calibration: Random-Initialisation Baseline

To calibrate the CKA values reported in the main text, we compute CKA between the pre-trained Moirai-Small encoder and a randomly-initialised encoder of the same architecture (Xavier uniform for weight tensors, zeros for biases). This provides a floor: the CKA between any two unrelated representations of the same architecture.

Table 5: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and CKA$\geq$0.979. Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small's negative forgetting. Mean $\pm$ std across 3 seeds (5 for r=8 Small); Moirai-Large single-seed (s=42) for $r \in \{16, 32, 64\}$, 3 seeds for r=8.

Backbone	Rank	h	MSE	Forg.%	CKA	Seeds
Small	4	96	.116$\pm$.001	-9.5 ± 0.9	.992$\pm$.001	3
Small	4	192	.146$\pm$.003	-8.8 ± 1.9	.995$\pm$.002	2
Small	8	96	.116$\pm$.001	-9.1 ± 1.1	.979$\pm$.006	5
Small	8	192	.157$\pm$.002	-1.8 ± 3.1	.986$\pm$.005	5
Small	16	96	.117$\pm$.001	-10.2 ± 1.2	.987$\pm$.002	3
Small	16	192	.155$\pm$.002	-1.7 ± 1.2	.988$\pm$.001	3
Small	32	96	.119$\pm$.001	-8.3 ± 0.9	.986$\pm$.001	3
Small	32	192	.159$\pm$.003	$+1.1\pm1.4$	.989$\pm$.002	3
Large	8	96	.159$\pm$.004	$+25.7\pm3.2$	.973$\pm$.012	3
Large	16	96	.154	+22.5	.955	1
Large	32	96	.150	+19.0	.986	1
Large	64	96	.165	+29.9	.984	1

Table 6: LoRA-Large HP ablation (ETTh2, r=8, n=500, h=96). Baseline (default LR=1e-4, α=16, modules = q,v,out) forg.= +22.5%, seed 42. Only LR=1e-5 rescues LoRA-Large (5 seeds); all other axes fail.

Axis	Setting	Val MSE	Forg.%	CKA	Seeds
LR	1e-4 (default)	.157	+22.5	.955	1
	5e-5	.146	+15.3	.982	1
	1e-5	**.115**$\pm$.001	$\mathbf{-8.5}\pm0.7$	**.989**$\pm$.003	**5**
α	8	.155	+21.9	.975	1
	16 (default)	.157	+22.5	.955	1
	32	.158	+24.3	.984	1
Modules	q,v,out (default)	.157	+22.5	.955	1
	q,k,v	.153	+20.4	.987	1
	q,k,v,out	.151	+19.0	.988	1

Using 50 benign probe samples and 5 random seeds (42, 123, 456, 789, 999), the random-initialisation CKA baseline is reported in Table 7.

The random-init CKA is $\leq 10^{-4}$ across 5 seeds (reported at 4 decimals as 0.0000 ± 0.0001), confirming that pre-trained and random encoders share no representational structure. This is consistent with the theoretical expectation: CKA between an orthogonally-initialised encoder and any fixed reference encoder has expectation asymptotically zero in the feature dimension. Since IoT fine-tuned CKA (0.31–0.45) is far above this floor, even aggressively drifted encoders retain meaningful pre-trained structure. ETT fine-tuned CKA (0.81–0.94) is near-ceiling, confirming substantial preservation. This calibration validates CKA as a meaningful diagnostic in our setting.

F Linear Probing: Functional Representation Diagnostic

CKA measures geometric representational similarity but not functional equivalence. We therefore supplement CKA with a *linear probe*: a Ridge regression head (α=1.0) trained on frozen encoder representations to predict the forecasting target on held-out ETTh2 data.

Protocol. For each (condition, seed) configuration, we extract mean-pooled encoder representations from the Moirai-Small encoder on the held-out validation and test sets (192-timestep windows, same

Table 7: CKA calibration. Random-init CKA (0.000 ± 0.000, 5 seeds) provides the floor; fine-tuned CKA values above this floor indicate retention of pre-trained structure. IoT fine-tuning (CKA 0.31–0.45) retains meaningful structure despite aggressive drift. ETT fine-tuning (CKA 0.81–0.94) preserves most pre-trained structure.

Comparison	CKA
Pre-trained vs. Random-init (baseline, 5 seeds)	0.0000 ± 0.0001
Pre-trained vs. ETT fine-tuned (B, h=96)	0.937–0.964
Pre-trained vs. IoT fine-tuned (C, 200/5, 10 seeds)	0.420 ± 0.026
Pre-trained vs. Frozen encoder (D)	1.000

format as evaluation). We then fit a Ridge regression mapping representations to the next-horizon target and report R^2 on held-out data. The probe training set uses n_{probe}=300 held-out examples; the remainder forms the test split used for MSE evaluation. ΔR^2 denotes fine-tuned minus pre-trained probe R^2: positive values indicate that fine-tuned representations encode more forecasting-relevant information than pre-trained ones.

Interpretation. Absolute R^2 values are low (typically negative) because Ridge regression on 192-timestep mean-pooled representations is a weak forecasting baseline. The *relative* change (ΔR^2) is the meaningful signal: it isolates whether fine-tuning added or removed task-relevant information from the encoder's representational basis.

Key result. The sample-size sweep (Table 3) shows ΔR^2 monotonically increasing with training data (Ridge: $+0.12$ at 500 samples $\rightarrow$ $+0.67 \pm 0.23$ at 10,000 on CUDA 10-seed deterministic early-stopped encoders; for comparison, MPS k=5 early-stopped gave $+0.56 \pm 0.14$ and final-epoch k=5 gave $+0.78 \pm 0.14$), while CKA decreases monotonically. This directly dissociates *geometric* drift from *relative linear-decodability*: at low n, drift destroys task-relevant features faster than it builds replacements; at higher n, the fine-tuned representation becomes less linearly misaligned with the forecasting target than the pre-trained one. We emphasise that $R^2(\text{FT})$ remains *absolutely negative* at every sampled $n \in \{500, 1\text{k}, 2\text{k}, 5\text{k}, 10\text{k}\}$ (ZS: $R^2 = -6.90$; CUDA k=10 early-stopped: -6.79 at n=500 $\rightarrow$ -6.24 ± 0.24 at n=10k). $\Delta R^2 > 0$ is therefore a *relative-improvement* signal, not a claim that fine-tuned representations linearly encode the target well in absolute terms; the primary task-outcome signal remains forg.< 0 at $n \geq 2$k. **Cross-dataset ΔR^2 comparability.** Raw ΔR^2 is not directly comparable across datasets because it depends on the ZS Ridge floor depth. On ETTm2 (n=10k, CUDA k=10), the ZS Ridge floor is $R^2(\text{ZS}) = -24.52$ — 3.5× deeper than ETTh2's -6.90 — which proportionally inflates the raw ΔR^2 (+5.43 on ETTm2 vs. +0.67 on ETTh2). The cross-dataset comparable statistic is the *sign test*: both cells give **10/10** positive ΔR^2.

Horizon-sweep probe: head-specificity of the ΔR^2 gain. Table 8 reports Ridge ΔR^2 across three probe heads on four early-stopped encoders at n=10k (the same encoders as the main-text MLP sweep). The gain is concentrated on the *trained* 96-step head: $\Delta R^2 = +0.116 \pm 0.153$ (k=4). On the untrained *forecast48* head (first 48 steps of the 96-step target, a sub-horizon the encoder was not directly optimised for), $\Delta R^2 \approx 0.000 \pm 0.163$ — the gain vanishes at the horizon boundary. On the untrained *delta1* head, a GBM probe gives $R^2(\text{PT}) = +0.059 > 0$, establishing a positive-floor reference, with $\Delta R^2 = -0.056 \pm 0.030$ (decodability *falls* on the untrained readout). The head-specificity pattern is consistent with the encoder drifting toward the fine-tuning objective rather than developing generally better representations (§5.5).

Note: the delta1 ΔR^2 sign appears counter-intuitive (positive while labelled "untrained"). The large positive delta1 ΔR^2 reflects that the ZS delta1 Ridge baseline is very negative ($R^2(\text{ZS}) \approx -2.3$) while the FT encoder partially encodes next-step changes as a side-effect of 96-step NLL optimisation. The diagnostic signal is the GBM result (§5.5): GBM gives $R^2(\text{PT}) = +0.059 > 0$ and $\Delta R^2 = -0.056$, yielding $R^2(\text{FT}) = +0.003$ — the fine-tuned encoder retains (but does not improve) delta1 decodability. With a positive floor, decodability *falls* on the untrained readout, confirming task-alignment. Note: $R^2(\text{PT})$ varies slightly across GBM probe depths (depth 4: $+0.062$, depth 6: $+0.059$, depth 8: $+0.054$) because the probe is re-fit independently at each depth; this reflects probe capacity, not

Table 8: Ridge ΔR^2 by probe head on early-stopped encoders (n=10k, seeds $\{303, 777, 888, 999\}$). "Forecast96" is the trained head; "Forecast48" and "Delta1" are untrained. The gain is head-specific to the trained objective.

Head	Trained?	Ridge ΔR^2 per seed			
		s303	s777	s888	s999
Forecast96 (96-step ahead)	✓	$+0.092$	-0.087	$+0.266$	$+0.192$
Forecast48 (sub-horizon)	×	$+0.061$	-0.111	-0.156	$+0.199$
Delta1 (next-step delta)	×	$+1.248$	$+0.757$	$+1.191$	$+0.980$
Mean ± std (forecast96)		$+0.116 \pm 0.153$			
Mean ± std (forecast48)		-0.001 ± 0.163			
Mean ± std (delta1)		$+1.044 \pm 0.223$			

encoder structure.

ETTm2 delta1 GBM reprobe. We ran the delta1/GBM probe on the five confirmed ETTm2 n=10k encoders (seeds 42/101/123/202/303) to check for a positive-floor + ΔR^2>0 cell on a second dataset. Result: R^2(ZS)=-0.030 (negative floor) and per-seed ΔR^2=$\{+0.025, -0.060, +0.026, +0.030, +0.016\}$ (4/5 positive, mean $+0.007 \pm 0.031$). *Substantive negative:* the only positive-floor head (ETTh2 delta1/GBM, R^2(ZS)=$+0.059$) gives ΔR^2=-0.056 — no positive-floor + ΔR^2>0 cell was found. The ΔR^2>0 signal on the trained 96-step head operates in a deeply-negative-R^2 regime; see Appendix H for probe noise-floor analysis.

Per-seed breakdown at n=10k (CUDA 10-seed). Ridge ΔR^2 (all positive; mean $+0.668$±0.226): 42:$+0.205$; 101:$+0.736$; 123:$+0.937$; 202:$+0.604$; 303:$+0.793$; 456:$+0.823$; 777:$+0.344$; 789:$+0.873$; 888:$+0.561$; 999:$+0.804$. Absolute R^2(FT): all in $[-6.70, -5.97]$ ($R^2_{\text{ZS}} = -6.904$). MLP k=5 (all positive; mean $+0.701$±0.324): 42:$+0.979$; 101:$+0.762$; 123:$+0.948$; 202:$+0.771$; 303:$+1.007$; 456:$+0.266$; 777:$+0.076$; 789:$+0.847$; 888:$+0.449$; 999:$+0.901$.

ETTm2 n=10k absolute R^2 per seed. The ETTm2 ZS Ridge floor (R^2(ZS)$= -24.52$, same pre-trained encoder across all seeds) is 3.5× deeper than ETTh2's -6.90, explaining the raw ΔR^2 magnitude difference. Per-seed absolute values for the five available CUDA seeds:

Seed	R^2(ZS)	R^2(FT)	ΔR^2	Forg.%
42	-24.52	-14.13	$+10.39$	-26.8
101	-24.52	-15.61	$+8.91$	-27.4
123	-24.52	-19.87	$+4.64$	-31.1
202	-24.52	-20.83	$+3.69$	-28.8
303	-24.52	-20.41	$+4.11$	-29.5
Mean (5 seeds)	-24.52	-18.17±3.08	$+6.35$±3.08	-28.7±1.6

All five seeds show $R^2(\text{FT}) > R^2(\text{ZS})$ (i.e., fine-tuned representations are *less* linearly misaligned with the ETTm2 forecasting target than pre-trained ones), and all show ΔR^2>0. The cross-dataset comparable statistic is the sign test: both ETTh2 (**10/10**) and ETTm2 (**5/5** seeds: 42, 101, 123, 202, 303) give unanimous ΔR^2>0.

MLP depth sweep on early-stopped encoders. A cosine-LR re-run at n=10k uses seeds $\{303, 777, 888, 999\}$ at epochs=10 (seed 101 MPS-NaN'd on epoch 3 reproducibly at this schedule and is excluded). The `-probe-mlp-layers` argument sweeps hidden-layer depth $k \in \{1, 2, 5\}$ on the early-stopped encoders (probe: hidden=64 per layer, α=10^{-3}, early stopping with validation_fraction=0.1, max_iter=500):

Table 9: Per-seed results at n=5,000 (ETTh2, h=96, condition B, 20 epochs). Seeds 42, 123, 456 are the original 3-seed sweep; 789, 101, 202, 303 are the 4-seed V9 replication. Forgetting is bimodal; ΔR^2 is uniformly positive.

Seed	MSE(FT)	Forg.%	CKA	ΔR^2	R^2(FT)
42	.132	+4.8	.531	+.67	−6.24
101	.120	−4.9	.571	+.75	−6.16
123	.124	−1.4	.544	+.81	−6.10
202	.128	+1.4	.461	+.15	−6.76
303	.150	+19.0	.525	+.52	−6.39
456	.153	+21.4	.711	+.20	−6.70
789	.134	+5.9	.480	+.40	−6.51
Mean±std	.134±.012	+6.6±10.0	.546±.082	+.50±.26	−6.41±.26

Seed	Ridge ΔR^2	MLP k=1	MLP k=2	MLP k=5
303	$+0.092$	$+1.306$	$+1.418$	$+0.540$
777	-0.087	$+1.136$	$+1.554$	$+0.759$
888	$+0.266$	$+2.004$	$+2.361$	$+1.034$
999	$+0.192$	$+1.816$	$+1.949$	$+0.908$
Mean ± std	$+0.116$±0.132	$+1.566$±0.356	$+1.821$±0.368	$+0.810$±0.184

The cosine-LR Ridge ΔR^2 is smaller than the epochs=20 run (+0.56±0.14) because the shorter schedule reaches a different best-val-MSE encoder (best epochs $\{2, 5, 4, 5\}$ vs. $\{2, 4, 8, 4, 3\}$), and Ridge is most sensitive to the specific linear basis. The non-monotonic k=1 → k=2 → k=5 trend reflects the deeper probe fitting the pre-trained encoder better (per-seed mean $R^2(\text{PT})_{k=5} = -6.93$ vs. $R^2(\text{PT})_{k=1} = -8.48$), shrinking the Δ from above, not degraded fine-tuned encoding. All four seeds are individually positive for k=5 MLP ($+0.54, +0.76, +1.03, +0.91$), preserving the cost–benefit gap. This run's CKA is 0.592 ± 0.142 (mean best epoch 4.0 ± 1.22, higher than the earlier run's 4.2 ± 2.3 because of the earlier LR annealing); forgetting is $-7.13 \pm 1.34\%$, tighter than the epochs=20 run's $-11.7 \pm 5.3\%$ and fully consistent with the forg.< 0 finding.

Crossover regime: per-seed breakdown at n=5,000. Table 9 reports all 7 seeds individually at n=5,000. Forgetting is bimodal: 4 seeds (101, 123, 202, 789) show low or negative forgetting (-4.9% to $+5.9\%$, mean $+0.7\%$) and 3 seeds (42, 303, 456) show high forgetting ($+4.8\%$ to $+21.4\%$, mean $+15.1\%$). All 7 seeds show positive ΔR^2 (+0.15 to +0.81, mean +0.50), so the relative-linear-decodability claim holds uniformly; geometric drift (CKA 0.46–0.71, mean 0.55) varies less than forgetting. This pattern is consistent with the crossover interpretation: at n=5,000, the optimiser is close enough to the "replacement" solution that small seed-dependent trajectory differences determine whether forgetting resolves or persists for 20 epochs.

F.1 n=5,000 per-epoch trajectory analysis

To support the mechanistic claim in §6, we plot the per-epoch training trajectories of all seven n=5,000 seeds, colour-coded by forgetting mode. Figure 2 shows val MSE and weight drift across training. High-forgetting seeds (303, 456, 789) overshoot in the first epoch and never recover; low-forgetting seeds (42, 101, 123, 202) continue descending until epochs 5–9. The epoch at which val MSE reaches its minimum separates the modes (0.7 ± 0.5 vs. 6.5 ± 2.1); mid-training weight drift (epochs 5–10) does not. Code: `scripts/analyse_n5k_trajectories.py`.

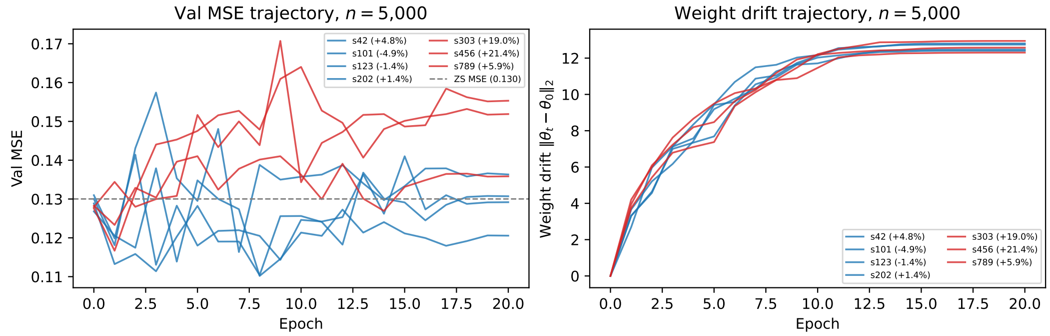

Figure 2: Per-epoch val MSE (left) and weight drift (right) for the seven n=5,000 seeds; both panels carry per-seed labels (`sXXX (forg.%)`) so each trajectory is individually identifiable. Blue: low-forgetting mode (forg. $\leq$ 5%); red: high-forgetting mode (forg. $>$ 5%). Grey dashed line: zero-shot MSE. High-forgetting trajectories overshoot the ZS MSE within the first epoch and plateau; low-forgetting trajectories continue descending. Mid-training weight drift (epochs 5–10) does not separate the modes.

G IoT Negative-Control Details

The IoT negative-control evaluation uses CICIoT2023 [21] with 12 network-flow features and 128-timestep windows; the anomaly score is the fraction of timesteps falling outside the 95% CI of Moirai's forecast over a 32-timestep scored horizon. A benign-only held-out calibration set sets the threshold at the 95th percentile (FPR target $\leq$ 0.05). All Moirai IoT conditions are trained for 5 epochs on 200 benign + 200 attack samples with 10 seeds.

CNN-NLL from-scratch control. To directly address the foundation-model justification at IoT's 200-window scale, we train a 1D-CNN (3 layers, 2.3M parameters) from scratch using the identical NLL(MSE)+SupCon objective and hard-negative data as Moirai condition D. At 10 seeds, CNN-NLL (AUC=0.598±0.041) matches fine-tuned Moirai B/D (AUC=0.556±0.021) and is close to the best Moirai condition C (AUC=0.629±0.022). A CNN-MSE variant (AUC=0.580±0.023) rules out the loss-function confound. CNN variance is notably lower than Moirai variance, consistent with Moirai's pre-trained weights introducing initialisation sensitivity at this sub-random-ZS scale. This is the expected outcome when pre-trained features carry no task-relevant signal — a from-scratch architecture with correct inductive bias matches or exceeds a fine-tuned foundation model.

H Probe Noise-Floor Analysis

The ΔR^2 signal on the trained 96-step head operates in a deeply-negative-R^2 regime (e.g., ZS -6.90, FT -6.24 at α=1). We test whether the ΔR^2>0 direction is an artefact of probe-design choices by running Ridge on two fixed encoders (ZS and condition-B seed 42, ETTh2, n=10k) across: (1) $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, (2) probe seeds $\{0, 1, 2\}$, (3) mean vs. last-token pooling. No retraining required (saved `best_encoder.pt`).

Results (Table 10).

Finding: ΔR^2>0 holds for all 10 probe-design settings tested; the direction is not an artefact of α, pooling, or probe seed. Magnitude decreases with stronger regularisation (larger α shrinks both R^2 values toward zero), but the sign is invariant. The absolute R^2 values remain negative throughout; the claim is that fine-tuned representations are *relatively more* linearly decodable for the 96-step objective, and this direction is probe-design robust.

Table 10: ΔR^2 under probe-design perturbations (ETTh2 96-step, ZS vs. FT seed 42). All $\Delta R^2 > 0$.

Setting	R^2(ZS)	R^2(FT)	ΔR^2
α=0.01	−13.51	−11.54	+1.96
α=0.1	−8.78	−7.94	+0.84
α=1.0 (default)	−6.90	−6.70	+0.20
α=10	−5.81	−5.72	+0.08
α=100	−5.32	−5.28	+0.04
Probe seeds 0/1/2 (α=1, mean pool)	same as α=1.0		+0.20
Last-token pool (α=1)	−9.70	−8.49	+1.20

I Random-Initialisation Negative Control

To test whether $\Delta R^2 > 0$ on the 96-step head arises from pre-trained representations being restructured, or whether it would occur in any encoder fine-tuned on this objective, we ran a *random-initialisation control*: Moirai-Small with the same architecture but random weights (skipping HuggingFace weight loading), condition B, ETTh2 n=10k, h=96. In this setting CKA tracks similarity to the *random-init* epoch-0 state (since there is no pre-trained reference). We ran 5 seeds (42/101/123/303/456) matching a subset of the condition B seed set.

Results (seeds 42/101/123/303/456):

Seed	CKA	R^2(PT)	R^2(FT)	ΔR^2	ZS MSE → FT MSE
42	0.357	−6.554	−6.692	−0.138	0.140 → 0.378
101	0.505	−6.780	−6.631	+0.149	0.164 → 0.398
123	0.540	−6.749	−6.944	−0.195	0.153 → 0.354
303	0.318	−6.834	−7.239	−0.405	0.151 → 0.359
456	0.273	−6.665	−7.070	−0.405	0.160 → 0.360
Mean±std	0.399±0.105			−0.199±0.205	

Across 5 seeds: $\Delta R^2 = -0.199 \pm 0.205$ (4/5 negative). The one positive seed (+0.149) is substantially smaller in magnitude than condition B's per-seed range (+0.38 to +0.75 for early-stopped seeds). None of the 5 random-init seeds approaches the pre-trained condition B mean (+0.668±0.226, 10/10 positive).

Compare to condition B (pre-trained): $\Delta R^2 = +0.363$ at seed 42 (and $+0.668 \pm 0.226$ across k=10 CUDA seeds). The random-init mean $\Delta R^2 = -0.199$ is substantially negative versus the pre-trained encoder's consistently positive signal. This confirms that the robust $\Delta R^2 > 0$ pattern is not a property of any encoder fine-tuned on the 96-step NLL objective; the pre-trained structure is required for the consistent positive directional shift. The decodability shift toward the objective reflects restructuring of pre-trained representations, not incidental training noise.

J Layer-Wise Unfreeze Localization

Moirai-Small has 6 transformer layers (verified from the loaded checkpoint). We added an `-unfreeze-top-n-layers` flag to `finetune_forecasting.py` and ran ETTh2 n=10k, h=96, 3 seeds (42/101/123) at N=3 alongside the existing N=0 (frozen encoder, 5-seed condition D) and N=6 (full unfreeze, condition B; N=6 unfreezes all 6 layers).

N=1 **instability.** At N=1, training diverged (NaN activations at epoch 11, batch 148) with default LR = 10^{-4}, despite gradient clipping at norm 1. The 11 epochs that completed show CKA stabilising at ≈0.72; we do not extract probe results from a crashed run. This instability is qualitatively consistent with the LoRA-Large behaviour documented in §5.4: extreme partial freeze concentrates gradient pressure on a small number of parameters, requiring LR reduction to stabilise.

N=3, N=4, N=5 results. Table 11 shows results for all six values of N: N=0 (5 seeds), N=3 and N=6 (3 matching seeds), and N=4 and N=5 (1 seed 42 each).

Table 11: Layer-wise unfreeze profile (ETTh2 n=10k, h=96; mean±std over seeds). N = number of top transformer layers unfrozen (Moirai-Small has 6 layers). N=0: 5 seeds; N=3/N=6: seeds 42/101/123; N=4/N=5: seed 42 only. Forgetting is final-epoch (no early stopping); N=5 overshoot reflects late-epoch divergence.

N	Layers	CKA	Ridge ΔR^2	Forgetting%
0 (frozen)	none	0.999±0.000	-0.048 ± 0.008	$-6.6\pm1.5\%$
1 (top-1)	1/6	≈0.72 (ep. 11)	*diverged (NaN)*	
3 (top-3)	3/6	0.698±0.066	$+0.479\pm0.115$ (**3/3**)	$+1.3\pm2.1\%$
4 (top-4)	4/6	0.610 (1 sd)	$+0.363$ (**1/1**)	$+4.1\%$
5 (top-5)	5/6	0.644 (1 sd)	$+0.767$ (**1/1**)	$+31.8\%^{\dagger}$
6 (full, B)	6/6	0.484±0.061	$+0.626\pm0.378$ (**3/3**)	$-7.7\pm8.4\%$

† Final-epoch overshoot; best-epoch (h=96, epoch 7) MSE=0.132≈ZS.

Finding: ΔR^2 and forgetting dissociate across unfreeze depth. $\Delta R^2>0$ appears already at N=3 (half the encoder, 3/3 seeds, CKA≈0.70), with substantially less geometric drift than full unfreeze (CKA≈0.48). N=4 and N=5 (1 seed each) both give $\Delta R^2>0$ (1/1), confirming the decodability signal extends through the full N=3–6 range. The ΔR^2 values at N=4 ($+0.363$) and N=5 ($+0.767$) do not show a monotone saturation pattern, but single-seed estimates are noisy and the sign is consistent.

The key dissociation is in *forgetting*: N=3 gives forg. $+1.3\pm2.1\%$ (task slightly worse than ZS), N=4 gives $+4.1\%$ (1 seed), and N=6 (full unfreeze, condition B) gives $-7.7\pm8.4\%$ (task substantially better). The N=5 final-epoch forgetting of $+31.8\%$ reflects a single seed that overshot (val MSE bottoms at epoch 7 before diverging); its best-epoch MSE is 0.132, essentially unchanged from ZS—qualitatively consistent with the N=5k trajectory analysis (§5.6). ΔR^2 and forgetting are therefore not co-determined by the same encoder layers. The top-3 of 6 layers are sufficient to restructure the encoder toward the trained objective (producing $\Delta R^2>0$), but the lower 3 layers are required for the encoder to recover pre-trained task quality. The additional ΔR^2 gain from N=3 to N=6 ($+0.48 \to +0.63$) is modest and has much higher variance ($0.115 \to 0.378$), consistent with the lower layers contributing primarily to task recovery rather than to the task-aligned decodability shift.

This finding sharpens the central claim: the frozen-encoder control (Appendix H) rules out data-distribution confound; the N=3 result rules out a whole-encoder-restructuring account. The decodability shift toward the objective is localised to the top transformer layers, and $\Delta R^2>0$ is a *necessary but not sufficient* predictor of task improvement across unfreeze levels. Both N=3 and N=6 operate in the same deeply-negative-R^2 regime; the probe-design robustness of Appendix H applies to both.

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